DSA Syllabus for Cohort 3.0

Complete Coding Journey – From Basics to Placement Level Mastery

- 350+ recorded video solutions covering important problems
- 25+ coding patterns covered (problem-solving focused)
- Advanced DSA concepts included
- 300 MCQs for concept clarity and practice
- 100+ assignment questions for hands-on coding
- One live session every week
- Mentor support on Discord
- Interview guidance
- Handwritten notes
- Online coding challenges
- Bonus Sheets:
  - Company-focused interview questions
  - Service-based & product-based question sets

BASICS OF PROGRAMMING

Concepts Covered

- Variables, Data Types
- Input/Output (Scanner, Print)
- Operators

- Expressions & Formulas
- Math API

Problem Table

#	Problem	Difficulty	Pattern
1	Sum of two integers	Easy	Input → Process → Output
2	Print message with values and sum	Easy	Expression Evaluation
3	Accept two integers and print sum	Easy	Input → Process → Output
4	Accept name, age and print formatted output	Easy	Input → Output Formatting
5	Increment & Decrement operator practice	Easy	Variable Manipulation
6	Swap two numbers (with & without 3rd variable)	Easy	Swap Pattern
7	Rectangle area & perimeter calculation	Easy	Formula Based
8	Compound Interest calculation	Easy	Mathematical Simulation
9	Triangle area using Heron's formula	Medium	Formula + Multi-step Calculation

CONTROL FLOW STATEMENTS

Concepts Covered

- if, if-else
- Nested conditions
- Ternary operator
- Type casting

Problem Table

#	Problem	Difficulty	Pattern
10	Greatest between two numbers	Easy	Comparison
11	Gender-based greeting message	Easy	Conditional Mapping
12	Handle wrong gender input + validation	Easy	Input Validation
13	Check even or odd	Easy	Modulo Check
14	Voter eligibility check (+ years left)	Medium	Conditional + Calculation
15	Day name from number (1–7)	Easy	Mapping / Conditional
16	Greatest among three numbers	Easy	Nested Comparison
17	Leap year check	Easy	Mathematical Condition
18	Shop discount calculation	Medium	Conditional + Formula
19	Electricity bill calculation	Medium	Slab Based Calculation

LOOPS & SWITCH STATEMENT

Concepts Covered

- for, while, do-while
- Nested loops
- break, continue
- switch statement
- Pattern printing (nested loops)

Problem Table

#	Problem	Difficulty	Pattern
20	Print "Hello World" n times	Easy	Iteration
21	Print natural numbers up to n	Easy	Iteration
22	Print n to 1 (reverse loop)	Easy	Reverse Iteration
23	Multiplication table of a number	Easy	Iteration + Formula
24	Sum up to n terms	Easy	Accumulation
25	Sum of even & odd numbers separately	Easy	Conditional + Accumulation
26	Print all factors of a number	Medium	Factor Pattern
27	Sum of all factors	Medium	Factor + Accumulation
28	Prime number check	Medium	Factor Pattern
29	Power of a number (a^b)	Medium	Repeated Multiplication
30	Print each digit of a number	Medium	Digit Extraction
31	Sum of digits	Medium	Digit Extraction + Accumulation
32	Reverse a number	Medium	Digit Manipulation
33	Palindrome number check	Medium	Digit Manipulation
34	Strong number check	Medium	Digit + Factorial
35	Automorphic number	Medium	Mathematical Pattern
36	Print "Hello" until wrong input (do-while)	Easy	Sentinel Loop
37	Menu-driven calculator (do-while)	Medium	Menu Loop
38	Print weekday using switch	Easy	Mapping (Switch)
39	Check vowel/consonant using switch	Easy	Character Mapping
40	Project - Guess Game	Medium	Simulation + Loop
41	Right Triangle - Star Pattern	Easy	Nested Loop Pattern

#	Problem	Difficulty	Pattern
42	Right Triangle - Number Pattern	Easy	Nested Loop Pattern
43	Right Triangle - Alphabet Pattern	Easy	Nested Loop Pattern
44	Inverted Right Triangle Pattern	Medium	Nested Loop Pattern
45	Mirrored Right Triangle Pattern	Medium	Nested Loop Pattern
46	Triangle Pattern	Medium	Nested Loop Pattern
47	V Pattern	Medium	Pattern Logic
48	X Pattern	Medium	Pattern Logic

METHODS AND ARGUMENTS

- Method Signature, Static & Non-Static Methods
- Factory Method
- Method Invocation using Object Reference

ARRAY DATA STRUCTURES (LEVEL - 1)

Concepts Covered

- Arrays (1D), indexing
- Traversal
- Reversal, Rotation
- Basic array operations

Problem Table

#	Problem	Difficulty	Pattern
49	Sum & average of array elements	Easy	Traversal + Accumulation
50	Find greatest element	Easy	Max Tracking
51	Find second greatest element	Medium	Max Tracking
52	Check if array is sorted	Easy	Traversal + Comparison
53	Reverse copy of array	Easy	Traversal + Reverse
54	Left rotation by 1	Easy	Rotation
55	Left rotation by K	Medium	Rotation
56	Reverse array without extra space	Easy	Two Pointer
57	Move negatives to left, positives to right	Medium	Partition / Two Pointer
58	Count subarrays with given sum	Medium	Prefix Sum

SEARCHING ALGORITHM

Concepts Covered

- Linear Search
- Binary Search

SORTING ALGORITHM

Concepts Covered

- Stable vs Unstable Sort
- Comparison-based sorting

- In-place sorting

Problem Table

#	Problem	Difficulty	Pattern
59	Bubble Sort	Easy	Comparison + Swapping
60	Insertion Sort	Easy	Insertion + Shifting
61	Selection Sort	Easy	Selection + Swapping

MATRIX (2D ARRAYS)

Concepts Covered

- 2D Arrays, Jagged Arrays
- Matrix traversal
- Index handling & exceptions

Problem Table

#	Problem	Difficulty	Pattern
62	Matrix Diagonal Sum	Easy	Traversal
63	Transpose Matrix	Easy	Index Transformation
64	Flipping an Image	Easy	Two Pointer (Row-wise)
65	Rotate Image	Medium	Transpose + Reverse
66	Search a 2D Matrix	Medium	Binary Search
67	Set Matrix Zeroes	Medium	Marking / In-place

TIME AND SPACE COMPLEXITY

Concepts Covered

- Time Complexity (Big-O)
- Space Complexity
- Complexity equations
- TLE (Time Limit Exceeded)
- Constraints & judge rules

OOPS INTRODUCTION

Concepts Covered

- Class, Object
- Constructor
- `this` keyword
- Encapsulation (basics)

Problem Table

#	Problem	Difficulty	Pattern
68	Design Private Cab Service	Medium	Object Modeling
69	Design Railway Ticket Class	Easy	Class Design

STRING

Concepts Covered

- String literals, immutability
- String methods
- Character operations
- Traversal & manipulation

Problem Table

#	Problem	Difficulty	Pattern
70	Print each character of string	Easy	Traversal
71	Reverse a string	Easy	Two Pointer
72	Palindrome using method (Two Pointer)	Easy	Two Pointer
73	Count vowels and consonants	Easy	Character Classification
74	Toggle each alphabet	Medium	Character Manipulation
75	Count strings with given prefix	Medium	Prefix Matching
76	Capitalize first & last character of each word	Medium	String Manipulation
77	Maximum number of words in sentences	Easy	String Parsing
78	Frequency of each character	Medium	Hashing
79	Check anagram strings	Medium	Hashing / Sorting
80	Sort words in a sentence	Medium	Sorting + Parsing
81	Rotate string	Medium	String Manipulation

COLLECTION FRAMEWORK

Concepts Covered

- Collections vs Collection
- ArrayList, HashSet, HashMap, TreeMap
- Dynamic data structures
- Hashing basics

Problem Table

#	Problem	Difficulty	Pattern
82	Print unique elements in array	Easy	HashSet
83	Frequency of elements in array	Easy	HashMap (Hashing)
84	Jewels and Stones	Easy	Hashing
85	Check if sentence is pangram	Easy	HashSet
86	First letter to appear twice	Easy	Hashing
87	Sum of unique elements	Easy	HashSet + Traversal
88	Sort the People	Medium	HashMap + Sorting
89	Most Frequent Even Element	Medium	HashMap + Frequency

Data Structures and Algorithmic Foundations

MEMORY AREAS & RECURSION - I

Concepts Covered

- Stack memory, stack frames
- Recursion basics
- Base case & recursive calls

Problem Table

#	Problem	Difficulty	Pattern
90	Print name N times	Easy	Recursion (Linear)
91	Print 1 to N	Easy	Recursion (Forward)
92	Print N to 1	Easy	Recursion (Backward)
93	Sum of first N numbers	Easy	Recursion (Accumulation)
94	Factorial of a number	Easy	Recursion (Multiplicative)

#	Problem	Difficulty	Pattern
95	Reverse an array	Easy	Recursion + Two Pointer
96	Check palindrome string	Easy	Recursion + Two Pointer
97	Fibonacci number	Easy	Recursion (Tree)

MATHEMATICS FOR PROGRAMMING

Concepts Covered

- GCD, LCM
- Prime numbers, factors
- Logarithmic operations
- Sieve technique
- Fast exponentiation

Problem Table

#	Problem	Difficulty	Pattern
98	GCD / HCF of two numbers	Easy	Euclidean Algorithm
99	GCD of array	Medium	GCD Reduction
100	LCM of two numbers	Easy	GCD + Formula
101	Count digits of a number	Easy	Logarithmic Trick
102	Median of data	Easy	Sorting / Logic
103	Palindromic number	Easy	Digit Manipulation
104	Factors of a number	Easy	Factor Pattern (√n)
105	Prime number check	Easy	Factor Pattern (√n)
106	Count primes (Sieve of Eratosthenes)	Medium	Sieve Technique
107	Kth factor of n	Medium	Factor Traversal

#	Problem	Difficulty	Pattern
108	Square root (Integer)	Medium	Binary Search
109	Power (x^n)	Medium	Fast Exponentiation
110	Power of Two	Easy	Bit Manipulation

ARRAYS & ARRAYLIST PROBLEMS (LEVEL - 2)

Concepts Covered

- ArrayList basics
- Two Pointer
- Prefix Sum
- Sliding Window
- Greedy / Kadane's

Problem Table

#	Problem	Difficulty	Pattern
111	Max Consecutive Ones	Easy	Sliding Window
112	Merge Sorted Array	Easy	Two Pointer
113	Remove Duplicates from Sorted Array	Easy	Two Pointer
114	Duplicate Zeros	Easy	Simulation
115	Valid Mountain Array	Easy	Traversal
116	Replace Elements with Greatest on Right	Easy	Reverse Traversal
117	Squares of a Sorted Array	Easy	Two Pointer
118	Move Zeroes	Easy	Two Pointer

#	Problem	Difficulty	Pattern
119	Sort Array By Parity	Easy	Partition / Two Pointer
120	Fibonacci Number	Easy	DP / Iteration
121	Two Sum	Easy	Hashing
122	Running Sum of 1D Array	Easy	Prefix Sum
123	Find Pivot Index	Easy	Prefix Sum
124	Product of Array Except Self	Medium	Prefix + Suffix
125	Maximum Subarray (Kadane's Algo)	Medium	Kadane
126	Majority Element (Boyer Moore)	Medium	Voting Algorithm
127	3Sum	Medium	Sorting + Two Pointer
128	Best Time to Buy & Sell Stock	Easy	Kadane / Greedy
129	Best Time to Buy & Sell Stock II	Medium	Greedy
130	Trapping Rain Water	Hard	Two Pointer
131	Container With Most Water	Medium	Two Pointer

SECTION: SORTING ALGORITHMS

Concepts Covered

- Merge Sort
- Quick Sort
- Cyclic Sort

Problem Table

#	Problem	Difficulty	Pattern
132	Missing Number	Easy	Cyclic Sort

#	Problem	Difficulty	Pattern
133	Find All Numbers Disappeared in an Array	Easy	Cyclic Sort
134	Set Mismatch	Easy	Cyclic Sort
135	Find All Duplicates in an Array	Medium	Cyclic Sort
136	Find the Duplicate Number	Medium	Cycle Detection
137	First Missing Positive	Hard	Cyclic Sort

BINARY SEARCH

Concepts Covered

- Binary Search (Iterative, Recursive)
- Search space reduction
- Binary Search on Answer

Problem Table

#	Problem	Difficulty	Pattern
138	Binary Search on descending array	Easy	Binary Search
139	Order Agnostic Binary Search	Easy	Binary Search
140	Guess Number Higher or Lower	Easy	Binary Search
141	Search Insert Position	Easy	Binary Search
142	Ceiling of a number	Easy	Binary Search
143	Floor of a number	Easy	Binary Search
144	Smallest letter greater than target	Easy	Binary Search
145	Valid Perfect Square	Easy	Binary Search
146	First Bad Version	Easy	Binary Search

#	Problem	Difficulty	Pattern
147	First and Last Position in Sorted Array	Medium	Binary Search
148	Count of element in sorted array	Medium	Binary Search
149	Peak in Mountain / Bitonic array	Medium	Binary Search
150	Search in Mountain / Bitonic array	Medium	Binary Search
151	Minimum in Rotated Sorted Array	Medium	Binary Search
152	Search in Rotated Sorted Array	Medium	Binary Search
153	Search in Infinite Sorted Array	Medium	Binary Search
154	Book Allocation Problem	Hard	Binary Search on Answer
155	Capacity to Ship Packages	Hard	Binary Search on Answer
156	Minimum Days to Make Bouquets	Hard	Binary Search on Answer
157	Smallest Divisor Given Threshold	Hard	Binary Search on Answer
158	Koko Eating Bananas	Hard	Binary Search on Answer
159	Split Array Largest Sum	Hard	Binary Search on Answer
160	Aggressive Cows	Hard	Binary Search on Answer

HASHING (LEVEL - 2)

Concepts Covered

- Hashing fundamentals

- Collision handling
- HashMap / HashSet usage

Problem Table

#	Problem	Difficulty	Pattern
161	Intersection of Two Arrays	Easy	HashSet
162	Intersection of Two Arrays II	Easy	HashMap
163	Subarray Sum Equals K (Positive)	Easy	Sliding Window
164	Subarray Sum Equals K	Medium	Prefix Sum + HashMap
165	Subarray with 0 Sum	Medium	Prefix Sum + HashSet
166	Longest Subarray with Sum K	Medium	Prefix Sum + HashMap
167	Contiguous Array	Medium	Prefix Sum + HashMap
168	Longest Consecutive Sequence	Medium	HashSet
169	Count distinct elements in every window	Medium	Sliding Window + HashMap
170	Insert Delete GetRandom O(1)	Hard	HashMap + Design

LINKED LIST

Concepts Covered

- Linked List implementation
- Dynamic memory allocation
- Types of Linked List
- Two Pointer technique

Problem Table

#	Problem	Difficulty	Pattern
171	Design Linked List	Easy	Implementation
172	Middle of Linked List	Easy	Two Pointer (Tortoise & Hare)
173	Merge Two Sorted Lists	Easy	Two Pointer
174	Linked List Cycle	Easy	Cycle Detection (Floyd's)
175	Linked List Cycle II	Medium	Cycle Detection
176	Reverse Linked List	Easy	Iterative / Recursion
177	Convert Binary LL to Integer	Easy	Traversal
178	Merge K Sorted Lists	Hard	Heap / Divide & Conquer
179	Remove Duplicates from Sorted List	Easy	Traversal
180	Remove Duplicates II	Medium	Traversal
181	Intersection of Two Linked Lists	Easy	Two Pointer
182	Remove Linked List Elements	Easy	Traversal
183	Palindrome Linked List	Easy	Two Pointer
184	Delete Node in Linked List	Easy	Pointer Manipulation
185	Reverse Nodes in K Group	Hard	Linked List Manipulation
186	Swap Nodes in Pairs	Medium	Pointer Manipulation
187	Remove Nth Node From End	Medium	Two Pointer
188	Add Two Numbers	Medium	Simulation
189	Copy List with Random Pointer	Hard	HashMap / Cloning
190	Flattening a Linked List	Hard	Recursion / Divide & Conquer

SECTION: STACK & QUEUE

Concepts Covered

- Stack, Queue, Deque
- Implementation using arrays/collections
- Expression evaluation
- Monotonic stack

Problem Table

#	Problem	Difficulty	Pattern
191	Implement Queue using Stacks	Easy	Simulation
192	Implement Stack using Queues	Easy	Simulation
193	Infix to Postfix Conversion	Medium	Stack (Expression)
194	Prefix to Infix Conversion	Medium	Stack (Expression)
195	Prefix to Postfix Conversion	Medium	Stack (Expression)
196	Postfix to Prefix Conversion	Medium	Stack (Expression)
197	Postfix to Infix Conversion	Easy	Stack (Expression)
198	Infix to Prefix Conversion	Medium	Stack (Expression)
199	Next Larger Element	Easy	Monotonic Stack
200	Smaller on Left	Easy	Monotonic Stack
201	Next Greater Element I	Easy	Monotonic Stack
202	Smallest number on left	Easy	Monotonic Stack
203	Next Smaller Element	Easy	Monotonic Stack
204	Backspace String Compare	Medium	Stack / Two Pointer
205	Stock Span Problem	Medium	Monotonic Stack
206	Online Stock Span	Medium	Monotonic Stack
207	Min Stack	Medium	Stack Design
208	Largest Rectangle in Histogram	Hard	Monotonic Stack

#	Problem	Difficulty	Pattern
209	Maximal Rectangle	Hard	Stack + Matrix
210	Celebrity Problem	Medium	Stack / Two Pointer
211	LRU Cache	Hard	HashMap + DLL
212	LFU Cache	Hard	HashMap + Design

Number System

Concepts Covered

- Bitwise operators
- Bit manipulation techniques

Problem Table

#	Problem	Difficulty	Pattern
213	Introduction to Bits and Tricks	Easy	Bit Basics
214	Check if i-th bit is set or not	Easy	Bit Masking
215	Check if number is odd or not	Easy	Bit Check
216	Check if number is power of 2	Easy	Bit Trick
217	Count number of set bits	Easy	Bit Counting
218	Set/Unset rightmost unset bit	Easy	Bit Manipulation
219	Swap two numbers	Easy	XOR Trick
220	Divide two numbers without * and /	Medium	Bit Manipulation
221	Minimum bit flips to convert number	Medium	XOR
222	Single Number - I	Medium	XOR
223	Power Set (Bit Manipulation)	Medium	Bit Masking
224	XOR of numbers in a range	Medium	Pattern + XOR

#	Problem	Difficulty	Pattern
226	Single Number - III	Medium	XOR + Bit Masking

SECTION: GREEDY APPROACH & PROBLEMS

Concepts Covered

- Greedy choice property
- Local vs global optimum
- Sorting + selection strategy

Problem Table

#	Problem	Difficulty	Pattern
227	Assign Cookies	Easy	Greedy + Sorting
228	Fractional Knapsack	Medium	Greedy (Ratio Based)
229	Lemonade Change	Easy	Greedy (Simulation)
230	Valid Parenthesis Checker	Hard	Stack (Validation)
231	N Meetings in One Room	Medium	Greedy (Activity Selection)
232	Jump Game I	Easy	Greedy
233	Jump Game II	Medium	Greedy
234	Minimum Platforms	Medium	Greedy + Sorting
235	Job Sequencing Problem	Medium	Greedy + Heap
236	Candy	Hard	Greedy
237	Shortest Job First	Medium	Greedy (Scheduling)
238	LRU Page Replacement	Medium	Greedy / Simulation
239	Insert Interval	Medium	Greedy + Intervals
240	Merge Intervals	Medium	Greedy + Sorting

RECURSION & BACKTRACKING - ADVANCED

Concepts Covered

- Tail & Head Recursion
- Backtracking
- Recursion tree

Problem Table

#	Problem	Difficulty	Pattern
241	K-th Symbol in Grammar	Medium	Recursion
242	Tower of Hanoi	Medium	Recursion
243	Josephus Problem	Medium	Recursion
244	Find Winner of Circular Game	Medium	Simulation / Recursion
245	Sort List / Stack (Recursive)	Medium	Recursion
246	Power Set	Medium	Backtracking
247	Subsets	Medium	Backtracking
248	Permutations	Medium	Backtracking
249	Subsets II	Medium	Backtracking
250	Letter Combinations of Phone Number	Medium	Backtracking
251	Combination Sum	Medium	Backtracking
252	Combination Sum III	Medium	Backtracking
253	Rat in a Maze	Medium	Backtracking
254	Sudoku Solver	Hard	Backtracking
255	N-Queens	Hard	Backtracking

#	Problem	Difficulty	Pattern
256	Word Search	Medium	Backtracking
257	Permutation Sequence	Hard	Backtracking + Math

BINARY TREE

Concepts Covered

- Tree basics & terminology
- Traversals (DFS, BFS)
- Recursive tree processing

Problem Table

#	Problem	Difficulty	Pattern
258	Inorder Traversal	Easy	DFS
259	Preorder Traversal	Easy	DFS
260	Postorder Traversal	Easy	DFS
261	Maximum Depth of Binary Tree	Easy	DFS
262	Level Order Traversal	Easy	BFS
263	Level Order Traversal II	Easy	BFS
264	Symmetric Tree	Easy	DFS
265	Path Sum	Easy	DFS
266	Populate Next Right Pointers	Medium	BFS
267	Diameter of Binary Tree	Medium	DFS
268	Balanced Binary Tree	Easy	DFS
269	Same Tree	Easy	DFS
270	Zigzag Level Order Traversal	Medium	BFS
271	Left View of Binary Tree	Medium	BFS/DFS
272	Right View of Binary Tree	Medium	BFS/DFS

#	Problem	Difficulty	Pattern
273	Top View of Binary Tree	Medium	BFS + Map
274	Bottom View of Binary Tree	Medium	BFS + Map
275	Vertical Order Traversal	Hard	BFS + Map
276	Sum of Longest Bloodline	Medium	DFS
277	Lowest Common Ancestor	Medium	DFS
278	Path Sum II	Medium	DFS + Backtracking
279	Path Sum III	Medium	Prefix Sum + DFS
280	Construct Tree (Preorder + Inorder)	Medium	Recursion
281	Burning Tree	Hard	BFS + Mapping
282	Morris Traversal	Hard	Threaded Tree

BINARY SEARCH TREE (BST)

Concepts Covered

- BST properties
- Search, insert, delete
- Inorder traversal (sorted property)

Problem Table

#	Problem	Difficulty	Pattern
283	Validate Binary Search Tree	Medium	DFS (Bounds Check)
284	Search in BST	Easy	BST Traversal
285	Kth Smallest in BST	Medium	Inorder Traversal
286	Floor and Ceil in BST	Medium	BST Traversal
287	Delete Node in BST	Medium	BST Manipulation

#	Problem	Difficulty	Pattern
288	Predecessor and Successor	Medium	BST Traversal
289	Lowest Common Ancestor in BST	Easy	BST Property
290	Two Sum IV (BST)	Easy	Hashing / Inorder
291	Convert Sorted Array to BST	Easy	Divide & Conquer

BALANCED TREES

- AVL Tree balancing and rotation
- Red-Black Tree

BINARY HEAP

Concepts Covered

- Min Heap, Max Heap
- Heapify, HeapSort
- PriorityQueue
- Comparator vs Comparable

Problem Table

#	Problem	Difficulty	Pattern
292	Last Stone Weight	Easy	Max Heap
293	Kth Largest Element in Array	Medium	Heap
294	K Closest Points to Origin	Medium	Heap
295	Top K Frequent Elements	Medium	Heap + Hashing
296	Top K Frequent Words	Medium	Heap + Sorting
297	Sort Characters By Frequency	Medium	Heap + Hashing

SLIDING WINDOW TECHNIQUE

Concepts Covered

- Fixed & variable window
- Two Pointer (expand/shrink)
- Window optimization

Problem Table

#	Problem	Difficulty	Pattern
298	Max Sum Subarray of size K	Easy	Fixed Window
299	Maximum Points You Can Obtain	Medium	Sliding Window
300	Count Occurrence of Anagram	Medium	Sliding Window + Hashing
301	Largest Subarray with Sum K	Medium	Variable Window
302	Pick Toys	Medium	Variable Window
303	Longest Substring Without Repeating Characters	Medium	Sliding Window + Hashing
304	Minimum Window Substring	Hard	Sliding Window + Hashing

GRAPH

Concepts Covered

- BFS, DFS
- Cycle detection
- Topological sort
- Shortest path & MST

Problem Table

#	Problem	Difficulty	Pattern
305	BFS Traversal	Easy	BFS
306	DFS Traversal	Easy	DFS
307	Cycle Detection (Undirected)	Medium	BFS/DFS
308	Cycle Detection (Directed)	Medium	DFS
309	Topological Sort (DFS)	Medium	DFS
310	Kahn's Algorithm	Medium	BFS
311	Topological Sort (Kahn's)	Medium	BFS
312	Cycle Detection (Kahn's)	Medium	BFS
313	Keys and Rooms	Medium	DFS
314	All Paths from Source to Target	Medium	DFS
315	Number of Islands	Medium	DFS/BFS
316	Dijkstra Algorithm	Medium	Shortest Path
317	Bellman Ford Algorithm	Medium	Shortest Path
318	Prim's Algorithm	Medium	MST
319	Kruskal Algorithm	Medium	MST
320	Floyd Warshall Algorithm	Hard	All-Pairs Shortest Path
321	Find City with Smallest Neighbors	Medium	Graph + Shortest Path
322	Flood Fill	Easy	BFS/DFS
323	Clone Graph	Medium	DFS/BFS
324	Number of Provinces	Medium	DFS/BFS
325	Course Schedule	Medium	Topological Sort
326	Cheapest Flights Within K Stops	Hard	BFS / Dijkstra

DYNAMIC PROGRAMMING (DP)

Concepts Covered

- Memoization, Tabulation
- Knapsack patterns
- String DP
- Partition DP

Problem Table

#	Problem	Difficulty	Pattern
327	Fibonacci Number	Easy	DP (1D)
328	Climbing Stairs	Easy	DP (1D)
329	Min Cost Climbing Stairs	Easy	DP (1D)
330	Jump Game	Medium	Greedy / DP
331	0/1 Knapsack	Medium	DP (Knapsack)
332	Subset Sum Problem	Medium	DP (Subset)
333	Partition Equal Subset Sum	Medium	DP (Subset)
334	Perfect Sum Problem	Medium	DP (Count Subsets)
335	Unbounded Knapsack	Medium	DP (Knapsack)
336	Rod Cutting	Medium	DP (Unbounded Knapsack)
337	Coin Change II	Medium	DP (Count Ways)
338	Coin Change	Medium	DP (Min Coins)
339	Longest Common Subsequence	Medium	DP (Strings)
340	Longest Common Substring	Medium	DP (Strings)
341	Print Longest Common Subsequence	Medium	DP (Reconstruction)
342	Shortest Common Supersequence	Medium	DP (Strings)

#	Problem	Difficulty	Pattern
343	Print Shortest Common Supersequence	Medium	DP (Reconstruction)
344	Longest Palindromic Subsequence	Medium	DP (Strings)
345	Minimum Deletions to Make String Palindrome	Medium	DP (Strings)
346	Matrix Chain Multiplication	Hard	DP (Partition)
347	Palindrome Partitioning	Hard	DP (Partition)
348	Longest Palindromic Substring	Medium	DP / Expand
349	Egg Dropping Puzzle	Hard	DP (Partition)
350	Scramble String	Hard	DP (Strings)

TRIE

Concepts Covered

- Trie structure
- Insertion, search, deletion
- Prefix-based operations

Problem Table

#	Problem	Difficulty	Pattern
351	Implement Trie (Prefix Tree)	Medium	Trie
352	Longest Common Prefix	Easy	Trie / String
353	Word Break	Medium	Trie + DP
354	Word Search II	Hard	Trie + Backtracking
355	Maximum XOR of Two Numbers in Array	Medium	Trie (Bit Manipulation)